

OCR Further Mathematics
Pure Core - Differential Equations
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) The displacement of a door from its equilibrium (closed) position is measured by the angle, θ radians, which the door makes with its closed position. The door can swing either side of the equilibrium position so that θ can take positive and negative values. The door is released from rest from an open position at time $t = 0$.

A proposed differential equation to model the motion of the door for $t \geq 0$ is

$$\frac{d^2\theta}{dt^2} + \lambda \frac{d\theta}{dt} + 3\theta = 0 \quad \text{where } m \text{ is a constant and } \lambda \geq 0.$$

- (i) According to the model, for what value of λ will the motion of the door be simple harmonic?

[1]

- (ii) Explain briefly why modelling the motion of the door as simple harmonic is unlikely to be realistic.

[1]

- (b) Find the range of values of λ for which the model predicts that the door will never pass through the equilibrium position.

[2]

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- (c) Sketch a possible graph of θ against t when λ lies outside the range found in part above but the motion is not simple harmonic.



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[1]

- 2(a) A particle P of mass 2 kg can only move along the straight line segment OA , where OA is on a rough horizontal surface. The particle is initially at rest at O and the distance OA is 0.9 m.

When the time is t seconds the displacement of P from O is x m and the velocity of P is v ms⁻¹. P is subject to a force of magnitude $4e^{-2t}$ N in the direction of A for any $t \geq 0$. The resistance to the motion of P is modelled as being proportional to v .

At the instant when $t = \ln 2$, $v = 0.5$ and the resultant force on P is 0 N.

Show that, according to the model, $\frac{dv}{dt} + v = 2e^{-2t}$. [3]

- (b) Find an expression for v in terms of t for $t \geq 0$. [5]

- (c) By considering the behaviour of v as t becomes large explain why, according to the model, P 's speed must reach a maximum value for some $t > 0$. [2]

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(d) Determine the maximum speed considered in part (c). [2]

(e) Determine the greatest value of t for which the model is valid. [4]



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3(a) A is a fixed point on a smooth horizontal surface. A particle P is initially held at A and released from rest.

It subsequently performs simple harmonic motion in a straight line on the surface. After its release it is next at rest after 0.2 seconds at point B whose displacement is 0.2 m from A . The point M is halfway between A and B .

The displacement of P from M at time t seconds after release is denoted by x m.

On the axes provided in the Printed Answer Booklet, sketch a graph of x against t for $0 \leq t \leq 0.4$.

[4]



(b) Find the displacement of P from M at 0.75 seconds after release.

[2]

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4(a) A particle of mass 2 kg moves along the x -axis. At time t seconds the velocity of the particle is $v \text{ ms}^{-1}$.

The particle is subject to two forces.

- One acts in the positive x -direction with magnitude $\frac{1}{2}t \text{ N}$.
- One acts in the negative x -direction with magnitude $v \text{ N}$.

Show that the motion of the particle can be modelled by the differential equation

$$\frac{dv}{dt} + \frac{1}{2}v = \frac{1}{4}t.$$



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[1]

(b) The particle is at rest when $t = 0$.

Find v in terms of t .

[5]



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(c) Find the velocity of the particle when $t = 2$.

[1]

(d)

When $t = 2$ the force acting in the positive x -direction is replaced by a constant force of magnitude $\frac{1}{2}$ N in the same direction.

Refine the differential equation given in the above part to model the motion for $t \geq 2$.

[1]

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(e) Use the refined model from the part above to find an exact expression for v in terms of t for $t \geq 2$.

[3]

- 5(a) A capacitor is an electrical component which stores charge. The value of the charge stored by the capacitor, in suitable units, is denoted by Q . The capacitor is placed in an electrical circuit.

At any time t seconds, where $t \geq 0$, Q can be modelled by the differential equation $\frac{d^2Q}{dt^2} - 2\frac{dQ}{dt} - 15Q = 0$.

Initially the charge is 100 units and it is given that Q tends to a finite limit as t tends to infinity.

Determine the charge on the capacitor when $t = 0.5$.

[6]



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- (b) Determine the finite limit of Q as t tends to infinity.

[1]

- 6(a) A particle of mass 0.5 kg is initially at point O. It moves from rest along the x-axis under the influence of two forces F_1 N and F_2 N which act parallel to the x-axis. At time t seconds the velocity of the particle is v ms⁻¹. F_1 is acting in the direction of motion of the particle and F_2 is resisting motion.

In an initial model

- F_1 is proportional to t with constant of proportionality $\lambda > 0$,
- F_2 is proportional to v with constant of proportionality $\mu > 0$.

Show that the motion of the particle can be modelled by the following differential equation. $\frac{1}{2} \frac{dv}{dt} = \lambda t - \mu v$ [2]

- (b) Solve the differential equation in part (a), giving the particular solution for v in terms of t , λ and μ . [7]

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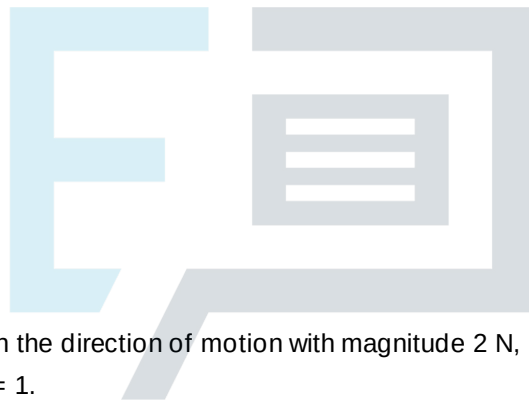
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(c) You are now given that $\lambda = 2$ and $\mu = 1$.

Find a formula for an approximation for v in terms of t when t is large.

[2]



(d) In a refined model

- F_1 is constant, acting in the direction of motion with magnitude 2 N,
- F_2 is as before with $\mu = 1$.

Write down a differential equation for the refined model.

[1]

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(e) Without solving the differential equation in part (d), write down what will happen to the velocity in the long term according to this refined model.

[1]

- 7(a) A particle is suspended in a resistive medium from one end of a light spring. The other end of the spring is attached to a point which is made to oscillate in a vertical line.

The displacement of the particle may be modelled by the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 10 \sin t$$

where x is the displacement of the particle below the equilibrium position at time t .

When $t = 0$ the particle is stationary and its displacement is 2.

Find the particular solution of the differential equation.

[11]



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(b) Write down an approximate equation for the displacement when t is large.

[2]



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- 8 In a predator-prey environment the population, at time t years, of predators is x and prey is y .
The populations of predators and prey are measured in hundreds.

The populations are modelled by the following simultaneous differential equations.

$$\frac{dx}{dt} = y \quad \frac{dy}{dt} = 2y - 5x$$

- (a) Show that $\frac{d^2x}{dt^2} = 2\frac{dx}{dt} - 5x$. [2]

- (b) (i) Find the general solution for x . [3]

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- (ii) Find the equivalent general solution for y . [2]

Initially there are 100 predators and 300 prey.

(c) Find the particular solutions for x and y .

[5]



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(d) Determine whether the model predicts that the predators will die out before the prey.

[4]



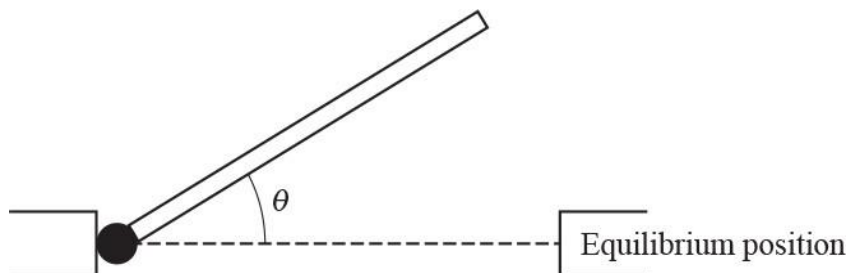
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- 9 A swing door is a door to a room which is closed when in equilibrium but which can be pushed open from either side and which can swing both ways, into or out of the room, and through the equilibrium position. The door is sprung so that when displaced from the equilibrium position it will swing back towards it.

The extent to which the door is open at any time, t seconds, is measured by the angle at the hinge, θ , which the plane of the door makes with the plane of the equilibrium position. See the diagram below.



In an initial model of the motion of a certain swing door it is suggested that it satisfies the following differential equation.

$$4 \frac{d^2\theta}{dt^2} + 25\theta = 0 \quad (*)$$

- (a) (i) Write down the general solution to (*).

[2]

- (ii) With reference to the behaviour of your solution in part (a)(i) explain briefly why the model using (*) is unlikely to be realistic. [1]

In an improved model of the motion of the door an extra term is introduced to the differential equation so that it becomes

$$4 \frac{d^2\theta}{dt^2} + \lambda \frac{d\theta}{dt} + 25\theta = 0 \quad (\dagger)$$

where λ is a positive constant.

- (b) In the case where $\lambda = 16$ the door is held open at an angle of 0.9 radians and then released from rest at time $t = 0$.

(i) Find, in a real form, the general solution of ($\dagger$). [3]

(ii) Find the particular solution of (†).

[4]



(iii) With reference to the behaviour of your solution found in part (b)(ii) explain briefly how the extra [2]
term in (†) improves the model.

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(c) Find the value of λ for which the door is critically damped.

[2]



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- 10 A particular radioactive substance decays over time.

A scientist models the amount of substance, x grams, at time t hours by the differential equation

$$\frac{dx}{dt} + \frac{1}{10}x = e^{-0.1t} \cos t.$$

- (a) Solve the differential equation to find the general solution for x in terms of t .

[3]



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Initially there was 10 g of the substance.

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- (b) Find the particular solution of the differential equation.

[2]

(c) Find to 6 significant figures the amount of substance that would be predicted by the model at

(i) 6 hours,

[1]

(ii) 6.25 hours.

[1]

(d) Comment on the appropriateness of the model for predicting the amount of substance over time.

[1]

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- 11 The quantity of grass on an island at time t years is x , in appropriate units. At time $t = 0$ some rabbits are introduced to the island. The population of rabbits on the island at time t years is y , in units of 100s of rabbits.

An ecologist who is studying the island suggests that the following pair of simultaneous first order differential equations can be used to model the population of rabbits and quantity of grass for $t \geq 0$.

$$\frac{dx}{dt} = 3x - 2y,$$

$$\frac{dy}{dt} = y + 5x$$

- (a) (i) Show that $\frac{d^2x}{dt^2} = a\frac{dx}{dt} + bx$ where a and b are constants which should be found.

[2]



- (ii) Find the general solution for x in real form.

[3]

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(b) Find the corresponding general solution for y .

[3]

At time $t = 0$ the quantity of grass on the island was 4 units. The number of rabbits introduced at this time was 500.

(c) Find the particular solutions for x and y .

[5]

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- (d) The ecologist finds that the model predicts that there will be no grass at time T , when there are still rabbits on the island.
Find the value of T .

[1]

- (e) State one way in which the model is not appropriate for modelling the quantity of grass and the population of rabbits for $0 \leq t \leq T$.

[1]

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- 12 In an experiment, at time t minutes there is Q grams of substance present.
It is known that the substance decays at a rate that is proportional to $1 + Q^2$. Initially there are 100 grams of the substance present and after 100 minutes there are 50 grams present.

Find the amount of the substance present after 400 minutes.

[8]



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- 13 One end of a light inextensible string is attached to a small mass. The other end is attached to a fixed point O . Initially the mass hangs at rest vertically below O . The mass is then pulled to one side with the string taut and released from rest. θ is the angle, in radians, that the string makes with the vertical through O at time t seconds and θ may be assumed to be small.

The subsequent motion of the mass can be modelled by the differential equation

$$\frac{d^2\theta}{dt^2} = -4\theta.$$

- (a) Write down the general solution to this differential equation. [1]

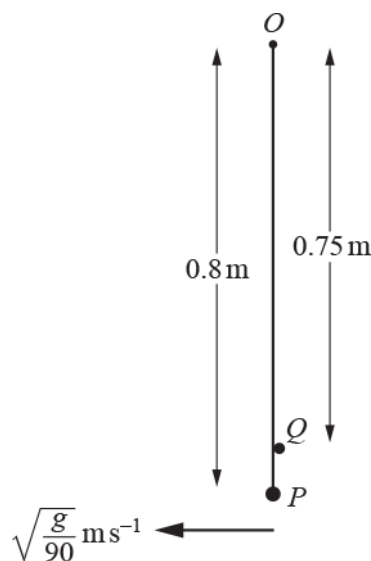
- (b) Initially the pendulum is released from rest at an angle of θ_0 .
Find the particular solution to the equation in this case. [4]

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- (c) State any limitations on the model. [1]



A light inextensible string of length 0.8 m is attached to a fixed point O . A particle P of mass m kg hangs in equilibrium attached to the lower end of the string. A small peg Q is fixed at a distance 0.75 m vertically below O .

The particle P is given a horizontal velocity of $\sqrt{\frac{g}{90}} \text{ ms}^{-1}$ so that the string initially moves to the left, away from the peg Q (see diagram).

- (i) Prove that the motion of P while it is to the left of its initial position is approximately simple harmonic, and find the period of this motion.

[7]

- (ii) Find the time that elapses between the first and second occasions that the string makes an angle of 5° with the vertical. Find also the linear speed of P in this position. [5]



- (iii) When the string returns to the vertical position, P begins to move in a circle with centre Q . Explain with a reason whether the motion of P is approximately simple harmonic motion while P is to the right of its initial position. [3]

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- 15 Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 25 \sin x. \quad [8]$$

- 16 Solve the differential equation

$$\frac{dy}{dx} + y \cot x = 9 \operatorname{cosec} x$$

to find y in terms of x subject to the condition $y = \pi$ when $x = \frac{1}{6}\pi$.

[8]

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- 17 A company is required to weigh any goods before exporting them overseas. When a crate is placed on a set of weighing scales, the mass displayed takes time to settle down to its final value.

The company wishes to model the mass, m kg, which is displayed t seconds after a crate X is placed on the scales.

For the displayed mass it is assumed that the rate of change of the quantity $\left(0.5 \frac{dm}{dt} + m\right)$ with respect to time is proportional to $(80 - m)$.

- (a) Show that $\frac{d^2m}{dt^2} + 2 \frac{dm}{dt} + 2km = 160k$, where k is a real constant. [2]

It is given that the complementary function for the differential equation in part (a) is $e^{\lambda t} (A \cos 2t + B \sin 2t)$, where A and B are arbitrary constants.

- (b) Show that $k = \frac{5}{2}$ and state the value of the constant λ . [4]

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When X is initially placed on the scales the displayed mass is zero and the rate of increase of the displayed mass is 160 kg s^{-1} .

- (c) Find m in terms of t . [7]

(d) Describe the long term behaviour of m . [1]

(e) With reference to your answer to part (d), comment on a limitation of the model. [1]

(f) (i) Find the value of m that corresponds to the stationary point on the curve $m = f(t)$ with the smallest positive value of t . [2]

(ii) Interpret this value of m in the context of the model. [1]

(g) Adapt the differential equation $\frac{d^2m}{dt^2} + 2\frac{dm}{dt} + 5m = 400$ to model the mass displayed t seconds after a crate Y, of mass 100 kg, is placed on the scales. [1]

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- 18 During an industrial process substance X is converted into substance Z . Some of the substance X goes through an intermediate phase, and is converted to substance Y , before being converted to substance Z . The situation is modelled by

$$\frac{dy}{dt} = 0.3x - 0.2y \quad \text{and} \quad \frac{dz}{dt} = 0.2y + 0.1x$$

where x , y and z are the amounts in kg of X , Y and Z at time t hours after the process starts.

Initially there is 10 kg of substance X and nothing of substances Y and Z . The amount of substance X decreases exponentially. The initial rate of decrease is 4 kg per hour.

- (a) Show that $x = Ae^{-0.4t}$, stating the value of A .

[3]



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- (b) (i) Show that $\frac{dx}{dt} + \frac{dy}{dt} + \frac{dz}{dt} = 0$.

[2]

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(ii) Comment on this result in the context of the industrial process.

[2]



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(c) Express y in terms of t .

[5]



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(d) Determine the maximum amount of substance Y present during the process.

[3]

(e) How long does it take to produce 9 kg of substance Z ?

[2]

19 Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 10y = 85 \cos x$$

20 Find the general solution of the differential equation

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$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = \sin x.$$



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- 21 Find the particular solution of the differential equation

$$x \frac{dy}{dx} + 3y = x^2 + x$$

for which $y = 1$ when $x = 1$, giving y in terms of x .

[8]

- 22 Solve the differential equation

$$\frac{d^2y}{dx^2} + 5 \frac{dy}{dx} + 6y = e^{-x}$$

subject to the conditions $y = \frac{dy}{dx} = 0$ when $x = 0$.

[10]

23

Find the solution of the differential equation $\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + 5y = e^{-x}$ for which $y = \frac{dy}{dx} = 0$ when $x = 0$.

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[11]

24

Solve the differential equation $x \frac{dy}{dx} - 3y = x^4 e^{2x}$ for y in terms of x , given that $y = 0$ when $x = 1$.

[8]



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25

The differential equation $\frac{d^2y}{dx^2} + 4y = \sin kx$ is to be solved, where k is a constant.

- (i) In the case $k = 2$, by using a particular integral of the form $ax \cos 2x + bx \sin 2x$, find the general solution.



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[7]

(ii) Describe briefly the behaviour of y when $x \rightarrow \infty$

[2]

(iii) In the case $k \neq 2$, explain whether y would exhibit the same behaviour as in part (ii) when $x \rightarrow \infty$

[2]



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